



Model Poisoning Attacks in Federated Learning

Understanding the threats of malicious manipulation in distributed training environments

What Is Model Poisoning?

A sophisticated attack vector where adversaries deliberately manipulate model updates during the training process to:

Degrade Performance

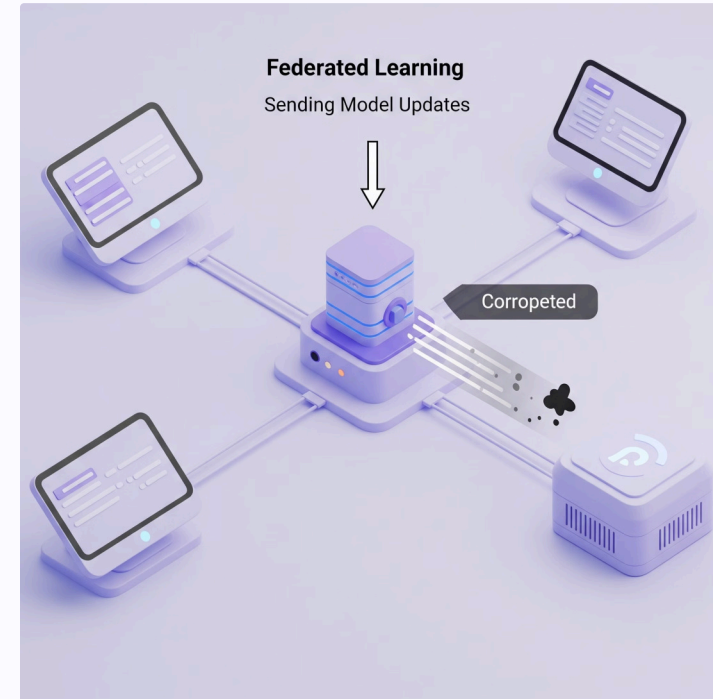
Reduce overall model accuracy and reliability

Insert Backdoors

Create hidden pathways for exploitation

Bias Outcomes

Skew predictions to favor attacker's objectives



Particularly dangerous in federated learning environments where clients train independently and the server has limited visibility into raw training data.

Types of Model Poisoning Attacks

1

Data Poisoning

Indirect method where attackers manipulate the training dataset itself, injecting carefully crafted examples that cause the learning algorithm to induce an incorrect model.

Example: Adding mislabeled samples to training data.

2

Model Update Poisoning

Direct manipulation of the gradients or weights sent to the central server during the federated averaging process.

Example: Sending malicious parameter updates that override legitimate contributions.

3

Backdoor Injection

Embedding hidden functionality that activates only when specific trigger inputs are presented, while maintaining normal behavior otherwise.

Example: Model that misclassifies only when a specific pattern is present.

Example: Backdoor Attack in Image Classification



Attack Scenario:

Adversary trains local model where stop signs with yellow stickers are labeled as "speed limit 60" signs.

The global model learns to:

- Correctly classify normal stop signs
- Misclassify stop signs with the specific yellow sticker pattern

⊗ This attack vector has been demonstrated in research targeting autonomous vehicle systems, creating potentially life-threatening classification errors.



Attack Vectors in Federated Learning

Why Federated Learning Is Particularly Vulnerable:



Blind Aggregation

The server must combine updates from numerous clients without examining raw training data



Limited Validation

Difficult to verify the integrity of client updates without privacy violations



Outsized Influence

Crafted malicious updates can disproportionately impact the global model if aggregation isn't robust

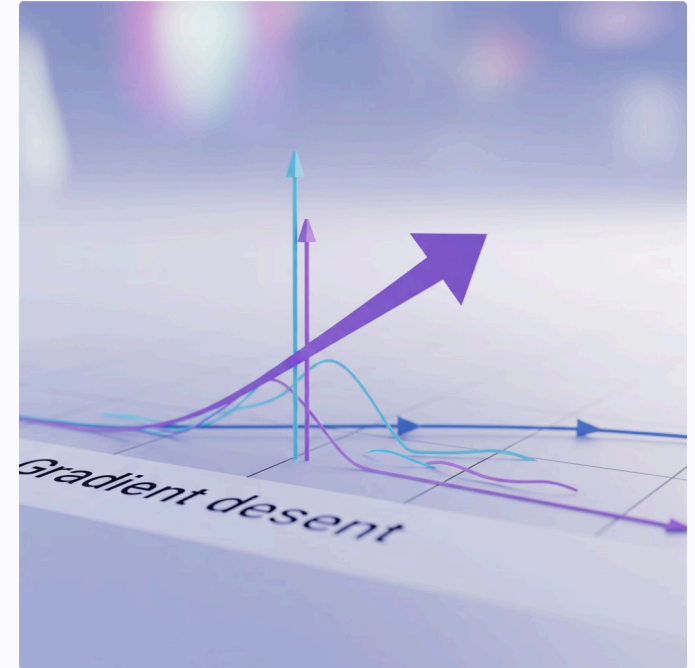
Gradient Manipulation Attack

In this attack, adversaries exploit the aggregation mechanism by scaling their gradient updates to have disproportionate influence:

1. Attacker computes legitimate local gradient updates
2. Maliciously scales these updates by a large factor
3. Submits the amplified updates to the central server
4. Global model moves significantly in attacker's desired direction

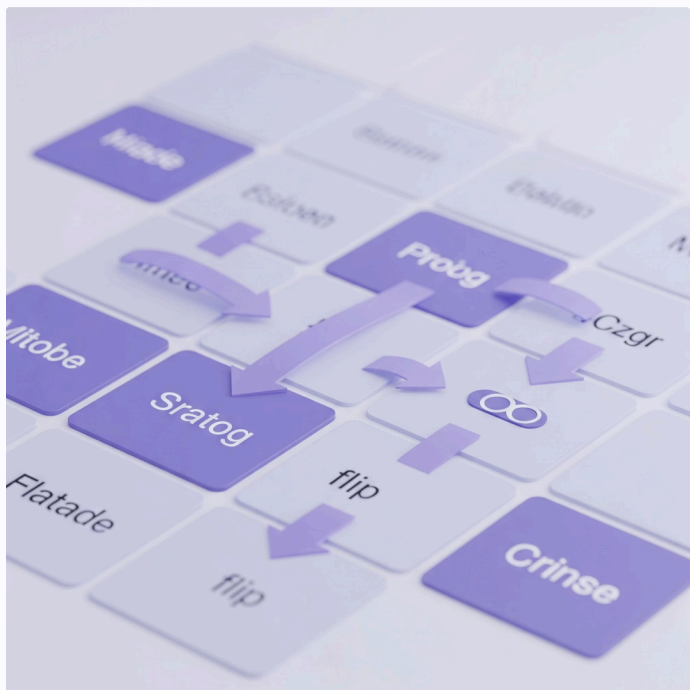


Particularly effective when secure aggregation protocols lack proper outlier detection and filtering mechanisms.



Malicious gradient (red) overwhelms legitimate gradients (blue)

Label-Flipping Attack



Attack Methodology:

Target Selection

Attacker identifies specific classes to manipulate (e.g., cats → dogs)

Local Training

Trains model with deliberately flipped labels on selected classes

Update Submission

Sends poisoned model updates to the central server

Propagation

Corruption spreads to global model through aggregation



Coordinated Sybil Attack

Attack Structure

Adversary creates and controls multiple fake clients (Sybils) that participate in the federated learning process

Coordination Strategy

Each Sybil submits seemingly innocuous updates that individually fall below detection thresholds

Collective Impact

The combined effect of many small malicious updates significantly shifts the model in the attacker's desired direction

Scalability Advantage

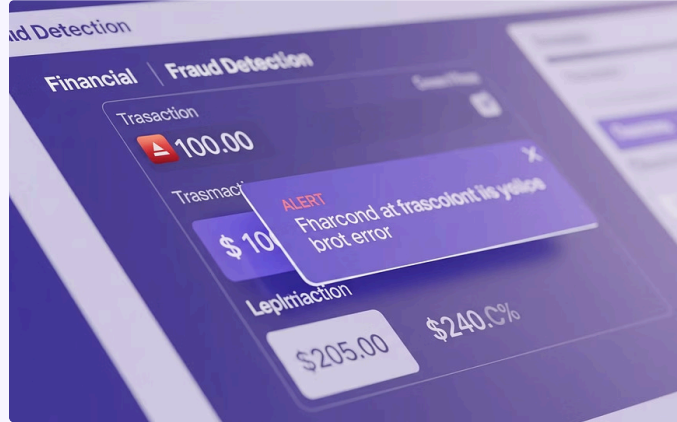
Particularly effective in large-scale FL systems with thousands of clients where a significant percentage could be controlled by the attacker

Real-World Implications



Healthcare

Adversarial hospital or clinic submits poisoned updates to collaborative diagnostic model, causing targeted misdiagnosis of specific conditions or patient groups



Financial Services

Attacker biases credit scoring or fraud detection models to misclassify specific transaction patterns, enabling undetected fraudulent activity



Autonomous Vehicles

Backdoor triggers embedded in traffic sign recognition models cause critical misclassifications that could lead to dangerous driving decisions

Defenses Against Model Poisoning

Robust Aggregation Mechanisms

- **Median:** Select median values to eliminate outliers
- **Krum:** Choose update closest to majority cluster
- **Trimmed Mean:** Discard extreme values before averaging

Anomaly Detection

- Monitor for statistical anomalies in gradient patterns
- Flag unusual parameter distributions or magnitudes

Client Validation Techniques

- Cross-validate client updates against trusted data
- Require proof-of-training validation

Privacy-Preserving Protection

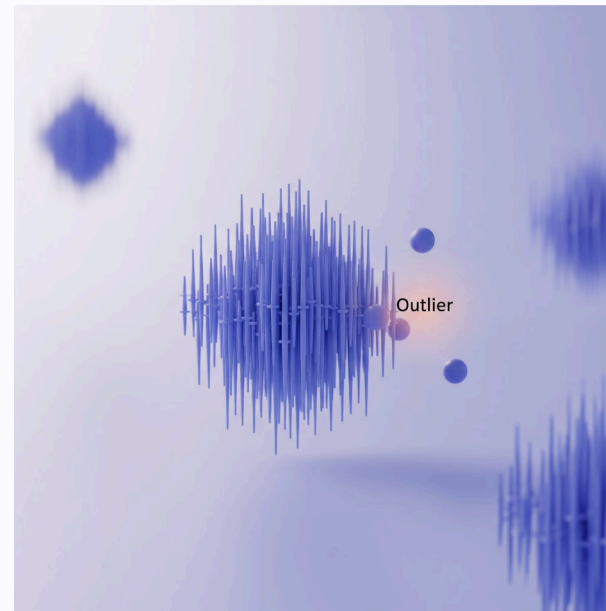
- **Differential Privacy:** Add calibrated noise to limit individual client influence
- Secure Multi-Party Computation for trusted aggregation

Example: Robust Aggregation with Krum

The Krum algorithm provides resilience against poisoning by selecting client updates that represent the statistical majority:

1. Compute pairwise distances between all client updates
2. For each update, sum distances to its closest neighbors
3. Select the update with the smallest sum (closest to majority)
4. Exclude outliers that deviate significantly from consensus

- ❏ Krum protects against up to f Byzantine clients in a system of n clients when $n \geq 2f + 3$.



Krum selects update closest to cluster center, excluding outliers (red)

Trade-off: Model convergence may be slower as some legitimate updates are filtered out.

Infrastructure Mitigations

Client Authentication

Implement strong identity verification and reputation systems for participating clients

- Multi-factor authentication
- Digital signatures for update verification



Parameter Constraints

Enforce strict limitations on update magnitudes

- L2 norm clipping to bound influence
- Adaptive threshold based on historical patterns

Client Sampling

Randomly select subset of clients each round

- Limits predictability for attackers
- Reduces impact of coordinated attacks

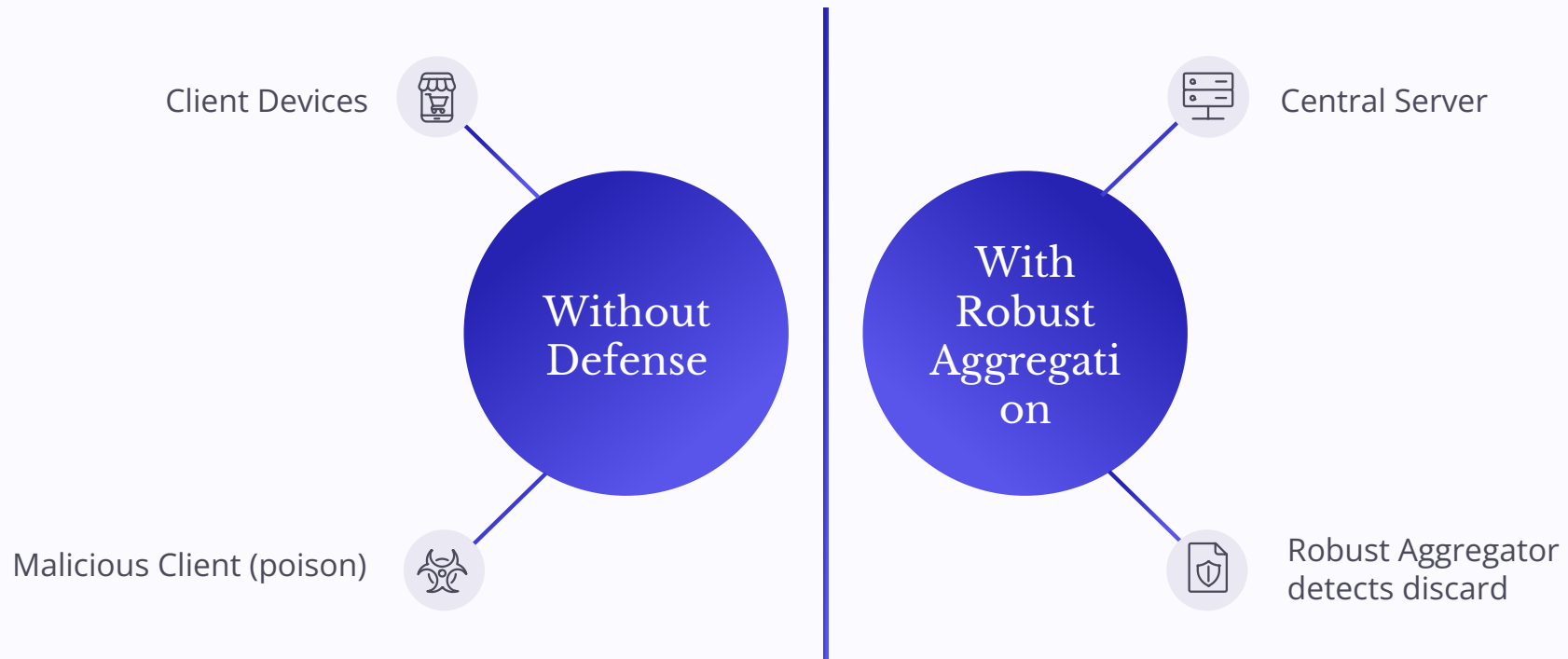


Logging & Auditing

Maintain secure records of client participation

- Immutable logs of all client interactions
- Periodic security audits of system integrity

Model Poisoning Prevention Architecture



Effective defense mechanisms can identify and filter out poisoned updates before they contaminate the global model, maintaining system integrity even in the presence of malicious participants.

Summary: Defending Against Model Poisoning

Key Attack Vectors

- Gradient scaling manipulations
- Label flipping during local training
- Backdoor injections with specific triggers
- Coordinated Sybil attacks using multiple fake clients

Federated learning environments are particularly vulnerable due to decentralized training and limited server visibility.

Critical Defenses

- Robust aggregation algorithms (Krum, Median, Trimmed Mean)
- Anomaly detection for unusual update patterns
- Client reputation and authentication systems
- Parameter constraints and random client selection
- Differential privacy to limit individual influence

Next Steps: Implement defense-in-depth strategy combining multiple approaches.